



DzNet Signal Coverage

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Difficulty Level : Medium

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A new telecom operator in Algeria wants to deploy its network across multiple cities. The cities are modeled as a graph where each city is a node, and connections between cities are given by an adjacency matrix.

The company can install a signal tower in any city. A tower installed in city i covers:
- city i itself, - every city directly connected to i .

A city may be covered by multiple towers, and some cities may have no connections.

Task: Determine the minimum number of signal towers required so that every city is covered.

Input

The first line contains an integer N ($1 \leq N \leq 20$) — the number of cities.

Each of the next N lines contains N space-separated integers (0 or 1) describing the adjacency matrix.

- 1 means there is a direct connection,
- 0 means there is no direct connection.

Output

Print one integer: the minimum number of towers needed to cover all cities.

Constraints

- $1 \leq N \leq 20$
- Matrix values are only 0 or 1
- Time limit: **2 seconds**
- Memory limit: **512 MB**

Sample Input 1

```
4
0 1 0 0
1 0 1 0
0 1 0 1
0 0 1 0
```

Sample Output 1

2

Sample Input 2

```
6
0 1 1 0 0 0
1 0 1 0 0 0
1 1 0 0 0 0
0 0 0 0 1 1
0 0 0 1 0 1
0 0 0 1 1 0
```

Sample Output 2

2

Sample Input 3

```
12
0 1 1 0 0 0 0 0 0 0 0 0
1 0 1 1 0 0 0 0 0 0 0 0
1 1 0 0 0 0 0 0 0 0 0 0
0 1 0 0 1 0 0 0 0 0 0 0
0 0 0 1 0 1 1 0 0 0 0 0
0 0 0 0 1 0 1 0 0 0 0 0
0 0 0 0 1 1 0 1 0 0 0 0
0 0 0 0 0 0 1 0 1 1 0 0
0 0 0 0 0 0 0 1 0 1 0 0
0 0 0 0 0 0 0 1 1 0 1 1
0 0 0 0 0 0 0 0 0 1 0 1
0 0 0 0 0 0 0 0 0 1 1 0
```

Sample Output 3

3

Explanation

In Sample 1, placing towers in cities 2 and 3 (1-based indexing) covers all four cities, and one tower alone is not enough.